University of Florida, Department of Mathematics

Analysis PhD Exam, May 1991

Be sure to give complete proofs and justify your steps. Do not leave gaps.

1. State:

(a) The Lebesgue Dominated Convergence Theorem

(b) Fatou's Lemma

(c) The Radon–Nikodym Theorem

(d) $(L^1(S,\Sigma,\mu))^* = L^\infty(S,\Sigma,\mu)$

2. State and prove the Vitali Convergence Theorem for integrals (convergence in measure form).

3. Show that a subset A of the reals is a set of Lebesgue measure zero if and only if there exists a sequence of intervals I_m such that

(a) $\displaystyle\sum_m \text{length}(I_m) < \infty$.

(b) $\displaystyle A \subset \bigcup_{m=k}^{\infty} I_m$ for every k.

4. Let f be Lebesgue integrable on $\mathbb{R}$. Suppose that $\displaystyle\int_0^x f\,dm = 0$ for every $x \in \mathbb{R}$. What can you conclude about f?

5. State and prove Fubini's Theorem.

6. Let $(S_m, \Sigma_m, P_m)_{m=1}^{\infty}$ be a sequence of probability spaces. Let $E_m \in \Sigma_m$, $m = 1, 2, \ldots$.

(a) Show that

$$E = \mathop{\times}_{m=1}^{\infty} E_m \in \bigotimes_{m=1}^{\infty} \Sigma_m.$$

(b) Prove that

$$\left(\bigotimes_{m=1}^{\infty} P_m\right)(E) = \prod_{m=1}^{\infty} P_m(E_m).$$